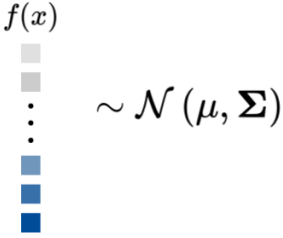
Introduction to Machine Learning

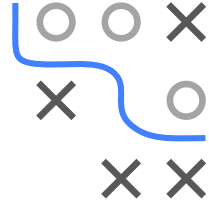
Gaussian Processes
Stochastic Processes and Distributions on Functions



Learning goals

- GPs model distributions over functions
- The marginalization property makes this distribution easily tractable
- GPs are fully specified by mean and covariance function
- GPs are indexed families

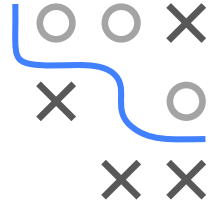
WEIGHT-SPACE VIEW



- Until now: hypothesis space $\mathcal{H}$ of parameterized functions $f(\mathbf{x} \mid \boldsymbol{\theta})$
- Bayesian paradigm: distribution over parameters (weights) $\boldsymbol{\theta}$
- Bayes' rule: update prior to posterior belief after observing data

$$p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) = \frac{\text{likelihood} \cdot \text{prior}}{\text{marginal likelihood}} = \frac{p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \cdot q(\boldsymbol{\theta})}{p(\mathbf{y}|\mathbf{X})}$$

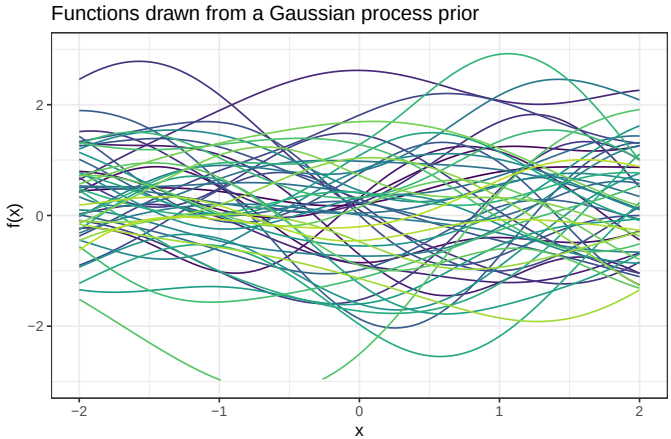
FUNCTION-SPACE VIEW



- New POV: rather than finding $\boldsymbol{\theta}$ to parameterize $f(\mathbf{x} \mid \boldsymbol{\theta})$ in parameter space, search space of admissible functions directly
- Sticking to Bayesian inference, specify prior distribution **over functions** and update according to observed data points

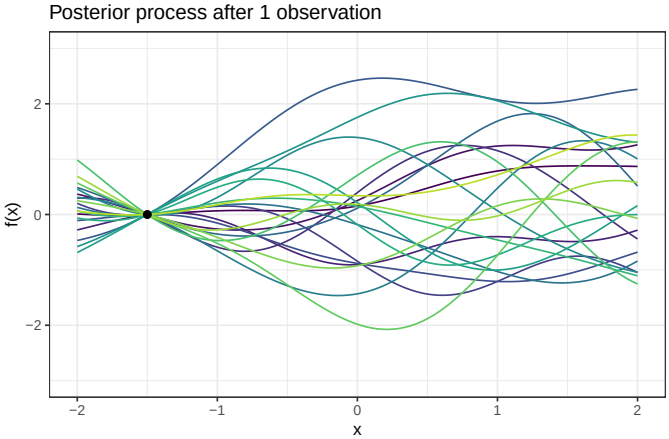
DRAWING FROM FUNCTION PRIORS

- Imagine we could draw functions from some prior distribution



DRAWING FROM FUNCTION PRIORS

- After observing some data points, we restrict sampling to functions consistent with the data



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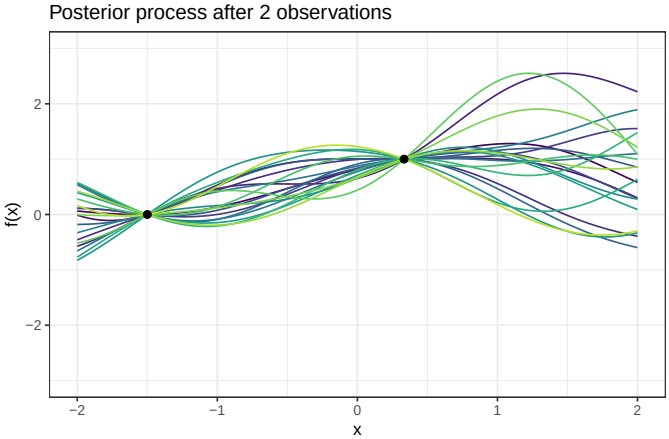

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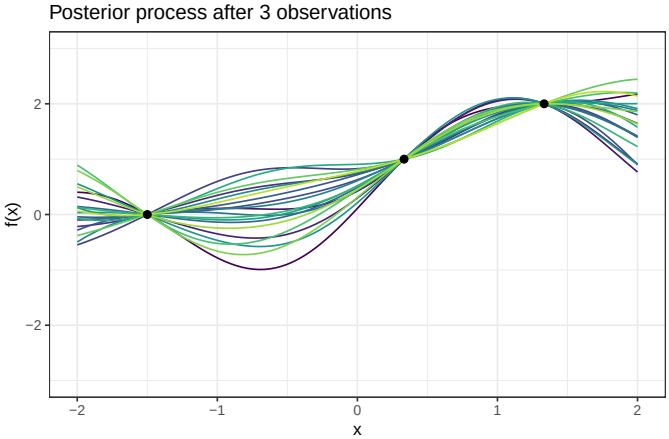
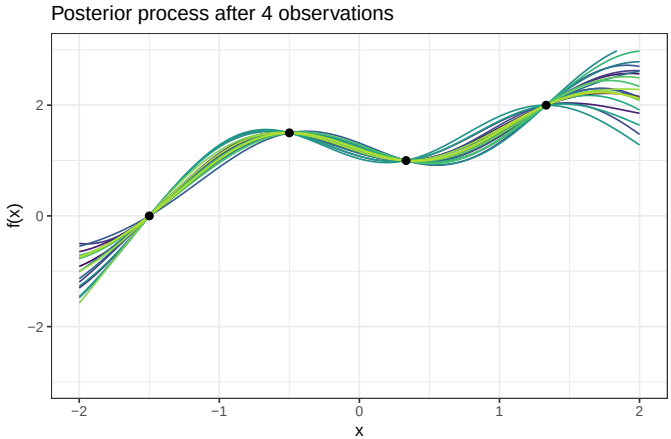


DRAWING FROM FUNCTION PRIORS

- Variety of admissible functions shrinks with observing more data



- Intuitively: distributions over functions have “mean” & “variance”

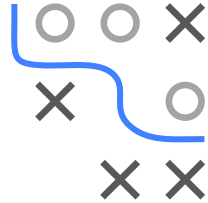
WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

Weight-Space View

- Parameterize functions (e.g., $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$)
- Define distributions on $\boldsymbol{\theta}$
- Inference in param space Θ

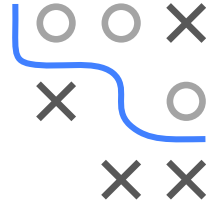
Function-Space View

- Work on functions directly
- Define distributions on f
- Inference in function space $\mathcal{H}$



Distributions on Functions

DISCRETE FUNCTIONS

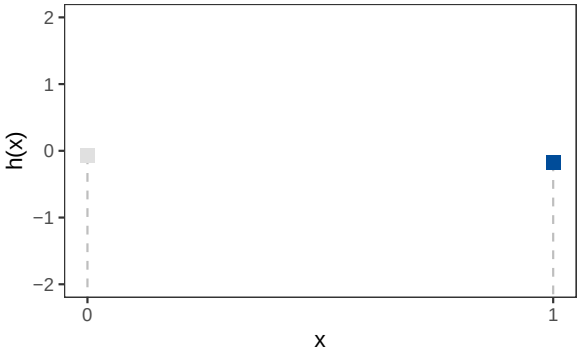


- Let $\mathcal{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$ be a finite set
- Let $\mathcal{H} = \{h \mid h : \mathcal{X} \to \mathbb{R}\}$
- Any $h \in \mathcal{H}$ has finite domain with $n < \infty$ elements $\Rightarrow$ neat representation with n-dim vector

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})]$$

RANDOM DISCRETE FUNCTIONS

- Case 1: input space $\mathcal{X} = \{0, 1\}$
- Example of functions living in this space



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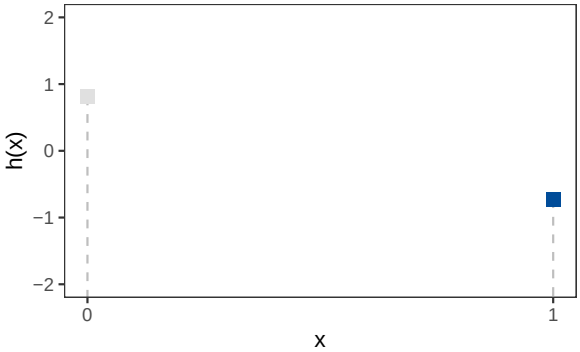

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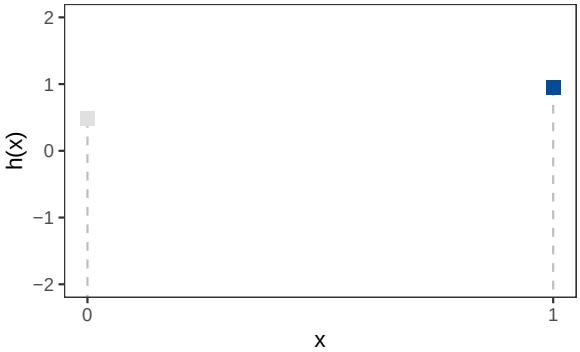

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- Case 2: input space $\mathcal{X} = \{0, 0.25, 0.5, 0.75, 1\}$
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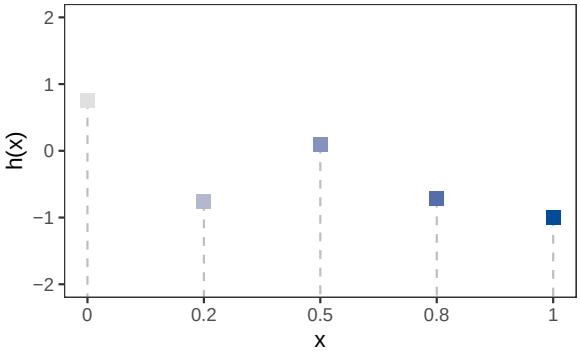

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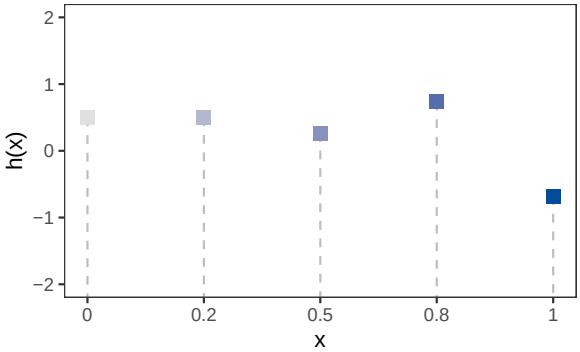

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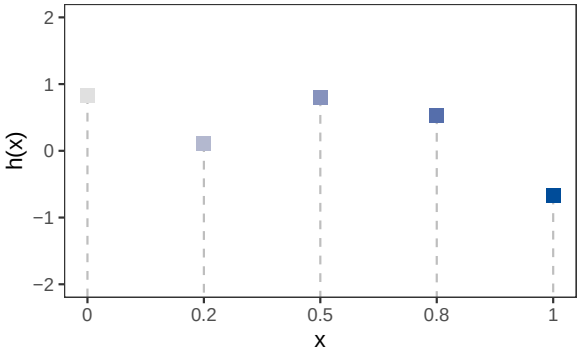

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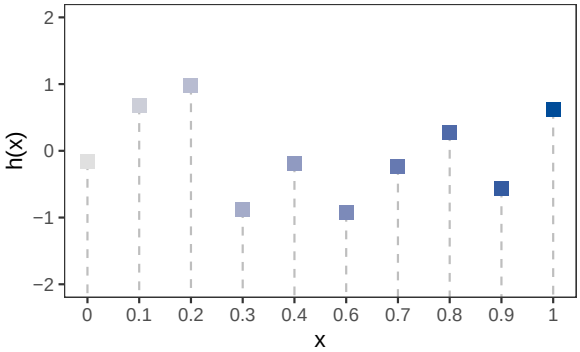
- Case 3: input space $|\mathcal{X}| = 10$
- Example of functions living in this space



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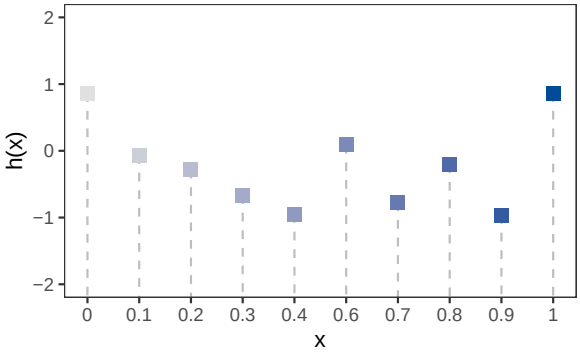

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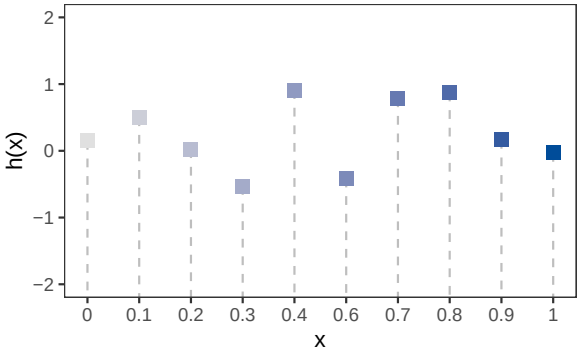
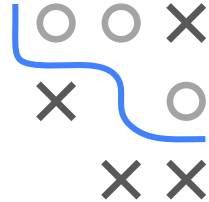


DISTRIBUTIONS ON DISCRETE FUNCTIONS



One natural way to specify a probability function on discrete function $h \in \mathcal{H}$ is to use the vector representation

$$\boldsymbol{h} = \left[h\left(\mathbf{x}^{(1)}\right), h\left(\mathbf{x}^{(2)}\right), \ldots, h\left(\mathbf{x}^{(n)}\right) \right]$$

of the function.

Let us see $\boldsymbol{h}$ as a n-dimensional random variable. We will further assume the following normal distribution:

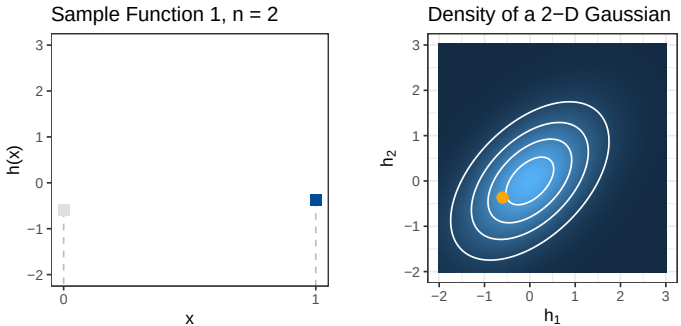
$$\boldsymbol{h} \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right).$$

Note: For now, we set $\boldsymbol{m} = \mathbf{0}$ and take the covariance matrix $\boldsymbol{K}$ as given. We will see later how they are chosen / estimated.

DISCRETE FUNCTIONS

Example 1 (continued): Let $h : \mathcal{X} \to \mathcal{Y}$ be a function that is defined on **two** points $\mathcal{X}$. We sample functions by sampling from a two-dimensional normal variable

$$\boldsymbol{h} = [h(1), h(2)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$



In this example, $m = (0, 0)$ and $K = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$.

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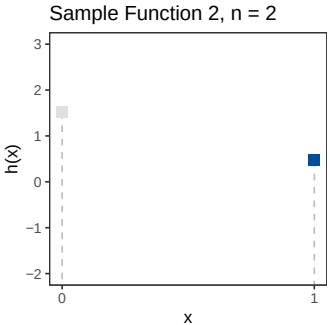

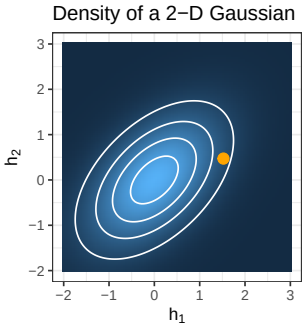

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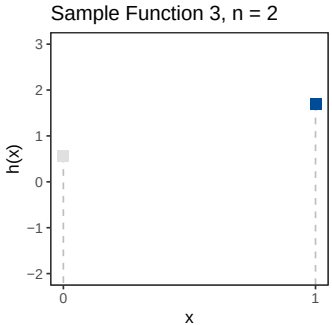

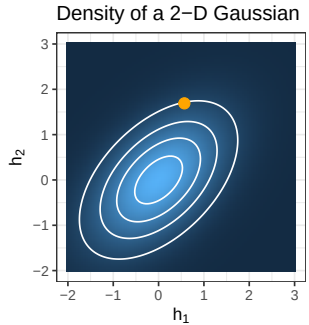

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Example 2 (continued): Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **five** points. We sample functions by sampling from a five-dimensional normal variable

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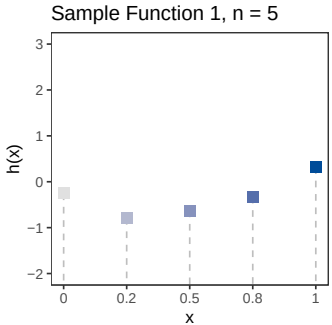

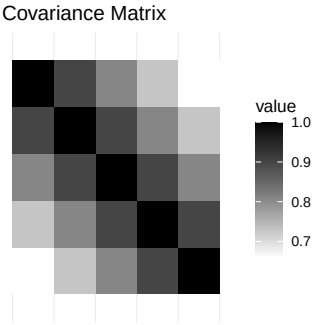

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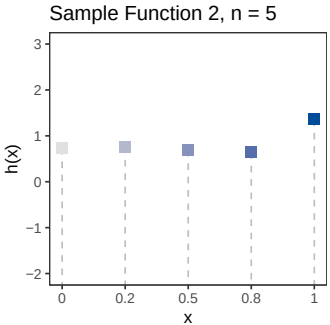

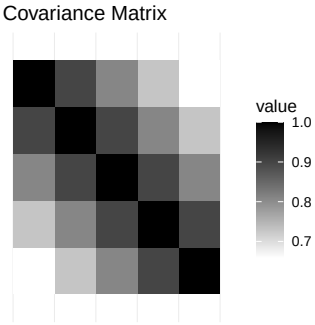

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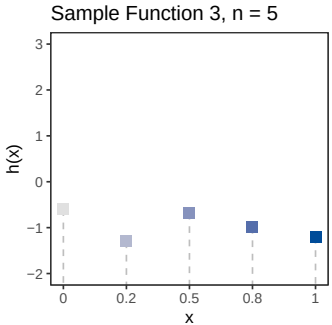

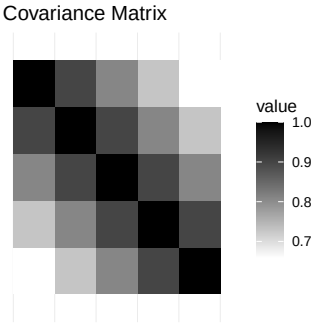

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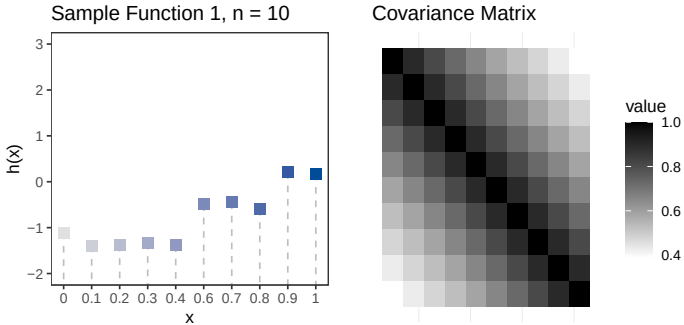

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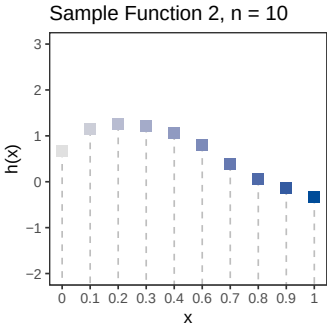

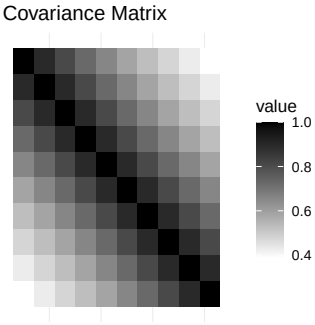

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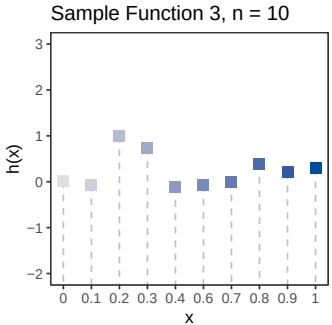
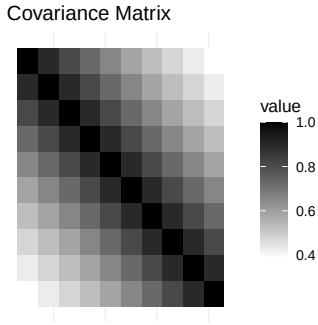


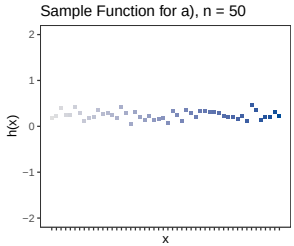


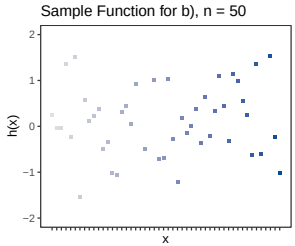
ROLE OF THE COVARIANCE FUNCTION

Note that the covariance controls the "shape" of the drawn function.
Consider two extreme cases where function values are

a) strongly correlated: $\boldsymbol{K} = \begin{pmatrix} 1 & 0.99 & \dots & 0.99 \\ 0.99 & 1 & \dots & 0.99 \\ 0.99 & 0.99 & \ddots & 0.99 \\ 0.99 & \dots & 0.99 & 1 \end{pmatrix}$

b) uncorrelated: $\boldsymbol{K} = \boldsymbol{I}$





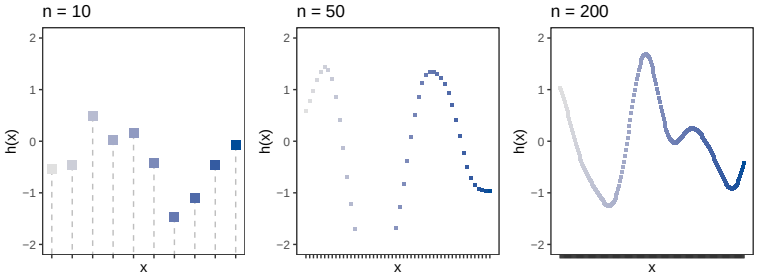
Gaussian Processes

FROM DISCRETE TO CONTINUOUS FUNCTIONS

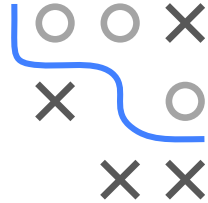
- We defined distributions on functions with discrete domain by defining a Gaussian on the vector of the respective function values

$$\mathbf{h} = [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)}), \dots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$

- We can do this for $n \to \infty$ (as "granular" as we want)



FROM DISCRETE TO CONTINUOUS FUNCTIONS



- No matter how large n is, we are still considering a function over a discrete domain.
- How can we extend our definition to functions with **continuous domain** $\mathcal{X} \subset \mathbb{R}$?

GAUSSIAN PROCESSES: INTUITION

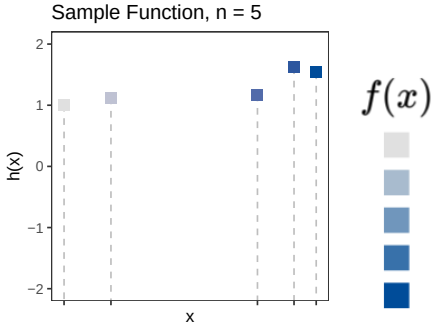
- Intuitively, a function f drawn from **Gaussian process** can be understood as an “infinite” long Gaussian random vector.
- It is unclear how to handle an “infinite” long Gaussian random vector!

GAUSSIAN PROCESSES: INTUITION

- Thus, it is required that for **any finite set** of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector $\mathbf{f}$ has a Gaussian distribution

$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

  with $\boldsymbol{m}$ and $\boldsymbol{K}$ being calculated by a mean function $m(.)$ / covariance function $k(.,.)$.
- This property is called **Marginalization Property**.



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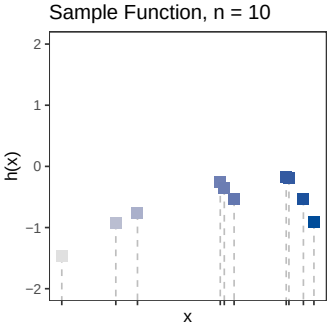

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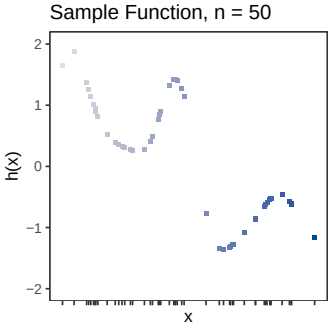

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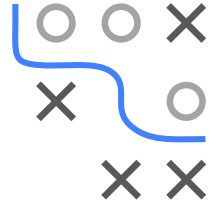
GAUSSIAN PROCESSES



This intuitive explanation is formally defined as follows:

A function $f(\mathbf{x})$ is generated by a GP $\mathcal{GP}\left(m(\mathbf{x}), k\left(\mathbf{x}, \mathbf{x}'\right)\right)$ if for **any finite** set of inputs $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$, the associated vector of function values $\boldsymbol{f} = \left(f(\mathbf{x}^{(1)}), \ldots, f(\mathbf{x}^{(n)})\right)$ has a Gaussian distribution

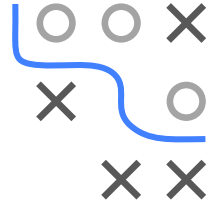
$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

with

$$\mathbf{m} \quad := \quad \left(m\left(\mathbf{x}^{(i)}\right)\right)_i, \quad \mathbf{K} := \left(k\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right)\right)_{i,j},$$

where $m(\mathbf{x})$ is called mean function and $k(\mathbf{x}, \mathbf{x}')$ is called covariance function.

SAMPLING FROM A GAUSSIAN PROCESS PRIOR



We can draw functions from a Gaussian process prior. Let us consider $f(\mathbf{x}) \sim \mathcal{GP}\left(0, k(\mathbf{x}, \mathbf{x}')\right)$ with the squared exponential covariance function [(*)]

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2\ell^2}\|\mathbf{x} - \mathbf{x}'\|^2\right), \quad \ell = 1.$$

This specifies the Gaussian process completely.

[(*)] We will talk later about different choices of covariance functions.

Gaussian Processes as Indexed Family

GAUSSIAN PROCESSES AS AN INDEXED FAMILY

A Gaussian process is a special case of a **stochastic process** which is defined as a collection of random variables indexed by some index set (also called an **indexed family**). What does it mean?

An **indexed family** is a mathematical function (or "rule") to map indices $t \in T$ to objects in $\mathcal{S}$.

Definition

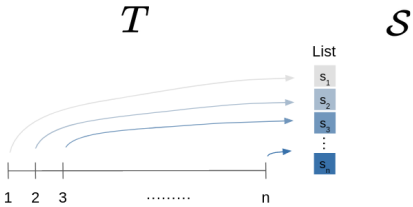
A **family of elements in $\mathcal{S}$ indexed by T** (indexed family) is a surjective function

$$\begin{aligned} s : T &\rightarrow \mathcal{S} \\ t &\mapsto s_t = s(t) \end{aligned}$$

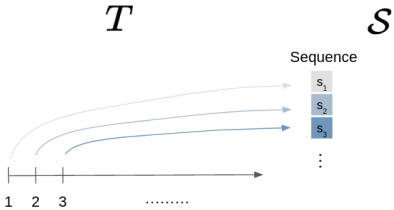
INDEXED FAMILY

Some simple examples for indexed families are:

- finite sequences (lists): $T = \{1, 2, \ldots, n\}$ and $(s_t)_{t \in T} \in \mathbb{R}$

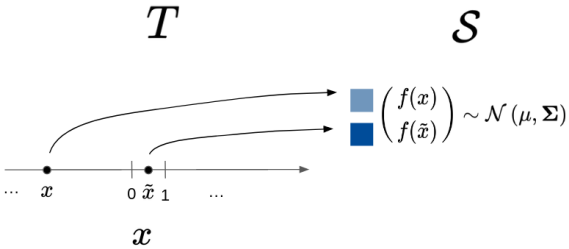


- infinite sequences: $T = \mathbb{N}$ and $(s_t)_{t \in T} \in \mathbb{R}$



INDEXED FAMILY

- A Gaussian process is also an indexed family, where the random variables $f(\mathbf{x})$ are indexed by the input values $\mathbf{x} \in \mathcal{X}$.
- Their special feature: Any indexed (finite) random vector has a multivariate Gaussian distribution (which comes with all the nice properties of Gaussianity!).



Visualization for a one-dimensional $\mathcal{X}$.

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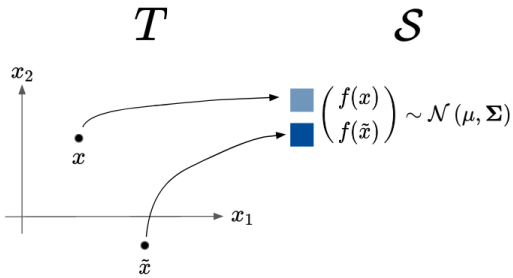

Visualization for a two-dimensional $\mathcal{X}$.